

LOCAL COMPONENT GROUPS AND THE ℓ -ADIC VALUATIONS OF THE MODULAR DEGREE

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ABSTRACT. Let $E/\mathbb{Q}$ be the $X_0(N)$ -optimal curve of its isogeny class and $\varphi_E: X_0(N) \rightarrow E$ its modular parametrisation. For a bad prime q let Φ_q^{geom} denote the component group of the Néron model of E over $\overline{\mathbb{F}}_q$. We study the inequality

$$v_\ell(\deg \varphi_E) \geq \sum_{q|N} v_\ell(\#\Phi_q^{\text{geom}})$$

for odd primes ℓ at which $E[\ell]$ is irreducible. For $\ell \geq 5$ its multiplicative part is a consequence of a theorem of Kim and Ota on the Pollack–Weston conjecture combined with a result of Agashe, Ribet and Stein, under hypotheses that exclude $\ell = 3$, primes $\ell \mid N$ and primes $q \equiv \pm 1 \pmod{\ell}$. We verify the inequality with no exception on every optimal curve of conductor at most 500000 (more than two million curves) and on several hundred curves of conductor up to $4 \cdot 10^6$ outside the tables, beyond all of these hypotheses. The new feature is the prime $\ell = 3$: every prime of Kodaira type IV or IV* contributes one to $v_3(\deg \varphi_E)$, independently of its Tamagawa number. We prove the qualitative statement by a level-lowering argument, formulate a quantitative conjecture with a supplement for the “twisted” types II, II* and I*_n, and describe the exact failure of the inequality in the Eisenstein case.

1. INTRODUCTION

Let $E/\mathbb{Q}$ be an elliptic curve of conductor N and let $f_E \in S_2(\Gamma_0(N))$ be the newform attached to it. Among the curves in the isogeny class of E there is a distinguished one, the $X_0(N)$ -optimal curve, for which the modular parametrisation $\varphi_E: X_0(N) \rightarrow E$ has minimal degree; throughout, E denotes this curve and $\deg \varphi_E$ its *modular degree*. The modular degree is computable [6, 22] and its growth is understood (it is $N^{1+o(1)}$ times a power of the discriminant), but its prime factorisation is not. The purpose of this note is to describe, and partly prove, a local-to-global divisibility: the ℓ -adic valuation of $\deg \varphi_E$ is bounded below by local data at the bad primes.

For a bad prime q let $\Phi_q^{\text{geom}} = \Phi_q(\overline{\mathbb{F}}_q)$ be the group of components of the special fibre of the Néron model of E at q , over the algebraic closure. Its order is read off the Kodaira type: n for type I_n; 1 for II, II*; 2 for III, III*; 3 for IV, IV*; 4 for I*_0 and I*_n [18, IV.9]. The Tamagawa number $c_q = \#\Phi_q(\mathbb{F}_q)$ counts only the components defined over $\mathbb{F}_q$; the statements below are about Φ_q^{geom} , not c_q .

Conjecture 1.1. *Let E be $X_0(N)$ -optimal and without complex multiplication, and let ℓ be an odd prime such that $E[\ell]$ is irreducible as a $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ -module. Then*

$$(1) \quad v_\ell(\deg \varphi_E) \geq \sum_{q|N} v_\ell(\#\Phi_q^{\text{geom}}).$$

For $\ell \geq 5$ only multiplicative primes contribute, through $v_\ell(n_q)$ where $n_q = v_q(\Delta_E)$ is the exponent of the minimal discriminant. For $\ell = 3$ the primes of type IV and IV* contribute one each. The multiplicative part has a long history. Ribet and Takahashi [17, 20] compared $\deg \varphi_E$ with the degree of a parametrisation by a Shimura curve and found the product of

the n_q over the primes of the quaternion discriminant, up to primes where $E[\ell]$ is reducible; Pasten [13] controls that error term. Pollack and Weston [14] conjectured, and Kim and Ota [10] proved, that moving a prime q into the “new” part of the level changes the congruence ideal of f_E by exactly the *Tamagawa exponent* t_q , the largest t with $\rho_{E,\ell} \bmod \ell^t$ unramified at q ; for multiplicative q one has $t_q = v_\ell(n_q)$ whether or not the reduction is split. Together with the theorem of Agashe, Ribet and Stein [1] that $\deg \varphi_E$ divides the congruence number r_E , with $v_p(\deg \varphi_E) = v_p(r_E)$ whenever $p^2 \nmid N$, this gives:

Theorem 1.2 (Kim–Ota, Agashe–Ribet–Stein). *Let $\ell \geq 5$ be a prime not dividing N such that $\bar{\rho}_{E,\ell}$ is absolutely irreducible when restricted to $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}(\sqrt{\ell^*}))$, $\ell^* = (-1)^{(\ell-1)/2}\ell$. Then*

$$v_\ell(\deg \varphi_E) \geq \sum_{\substack{q \parallel N \\ q \not\equiv \pm 1 \pmod{\ell}}} v_\ell(n_q).$$

The data presented in Section 4 say that none of the three restrictions in Theorem 1.2 is needed for the inequality: it holds at $\ell = 3$, at primes ℓ dividing N , and with the primes $q \equiv \pm 1 \pmod{\ell}$ included, in every one of more than two million optimal curves. The additive term at $\ell = 3$ is of a different nature, since $\bar{\rho}_{E,3}$ is ramified at a prime of type IV or IV* and no Tamagawa exponent is available. What makes it work is that the ramification is unipotent, so that the level can be lowered from q^2 to q :

Theorem 1.3. *Let $E/\mathbb{Q}$ have additive reduction of Kodaira type IV or IV* at a prime $q \geq 5$. Assume that $\bar{\rho}_{E,3}$ restricted to $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ is absolutely irreducible, and that $9 \nmid N$. Then 3 divides the modular degree of the $X_0(N)$ -optimal curve in the isogeny class of E .*

Theorem 1.3 is the qualitative half of the additive term; the quantitative half, namely that the contributions of several such primes and of the multiplicative primes add up, is Conjecture 1.1, and the data support it also at $q = 3$ and when $9 \mid N$.

Two further statements are supported by the data with no exception. The first concerns the *twisted* types. Quadratic twisting at an odd prime q exchanges $\text{II} \leftrightarrow \text{IV}^*$, $\text{IV} \leftrightarrow \text{II}^*$ and $\text{I}_n \leftrightarrow \text{I}_n^*$; at a prime of type II, II* or I_n* the mod 3 representation is a ramified quadratic character times a unipotent, that is, it has the shape of a twisted Steinberg representation. Give such a prime the weight $w_q = 1$ for types II, II* and $w_q = v_3(n_q)$ for type I_n*.

Conjecture 1.4. *With the hypotheses of Conjecture 1.1 and $\ell = 3$,*

$$v_3(\deg \varphi_E) \geq \sum_{q \mid N} v_3(\#\Phi_q^{\text{geom}}) + \left(\sum_q w_q - \max_q w_q \right).$$

In words, the twisted primes contribute all but the largest of their weights: two primes of type II force $3 \mid \deg \varphi_E$, one does not, three force $9 \mid \deg \varphi_E$. We have no mechanism for this statement.

The second statement concerns the Eisenstein case, in which Conjecture 1.1 can fail.

Conjecture 1.5. *Let E be optimal, not CM, with a rational ℓ -isogeny, ℓ odd, and let D be the deficit $\sum_q v_\ell(\#\Phi_q^{\text{geom}}) - v_\ell(\deg \varphi_E)$. Then*

$$D \leq v_\ell(\#E(\mathbb{Q})_{\text{tors}}) + [E[3] \simeq \mathbb{Z}/3 \oplus \mu_3],$$

and also $D \leq$ the number of curves in the isogeny class of E with a rational point of order ℓ . In particular $D \leq 0$ when E has no rational ℓ -torsion point, $D \leq 1$ for $\ell \geq 5$, and $D \leq 2$ for $\ell = 3$.

Section 2 collects the local facts, Section 3 proves Theorem 1.3, Section 4 describes the computations and Section 5 discusses the prime 2, where the analogous statements overlap with the literature on Watkins’s conjecture.

2. LOCAL INVARIANTS

Let q be a prime of bad reduction. We use Tate's algorithm [18, IV.9] for the Kodaira type and c_q , and the following two facts.

Lemma 2.1. *Let q be a prime of multiplicative reduction, $n_q = v_q(\Delta_E)$, and $\ell \neq q$ a prime. Then $\rho_{E,\ell} \bmod \ell^t$ is unramified at q if and only if $\ell^t \mid n_q$. This holds for split and for nonsplit reduction.*

Proof. Over the unramified quadratic extension of $\mathbb{Q}_q$ (which does not change the inertia group) E is a Tate curve with parameter u , $v_q(u) = n_q$. The ℓ^t -torsion is generated by μ_{ℓ^t} and an ℓ^t -th root of u , so inertia acts trivially on $E[\ell^t]$ exactly when u is an ℓ^t -th power in $\mathbb{Q}_q^{\text{ur}}$ up to units, that is, when $\ell^t \mid n_q$. $\square$

Lemma 2.2. *Let $q \geq 5$ be a prime at which E has reduction of type IV or IV*. Then the image of the inertia group I_q in $\text{Aut}(T_3E)$ is cyclic of order 3, and its image in $\text{Aut}(E[3])$ is a nontrivial unipotent subgroup. Consequently $E[3]^{I_q}$ is a line, $\bar{\rho}_{E,3}$ is tamely ramified at q , and the Serre conductor of $\bar{\rho}_{E,3}$ has exponent 1 at q .*

Proof. For $q \geq 5$ the reduction is potentially good and the semistability defect, the order of the image of I_q in $\text{Aut}(T_\ell E)$ for any $\ell \neq q$, equals $12/\gcd(v_q(\Delta_E), 12)$ [11]; for $v_q(\Delta_E) \in \{4, 8\}$ this is 3. The image is tame since $q \neq 3$, hence cyclic, generated by an element σ of order 3 with determinant 1, so with eigenvalues the two primitive cube roots of unity. Hence $\sigma^2 + \sigma + 1$ annihilates $T = T_3E$, and T is a module over $\mathbb{Z}_3[\sigma]/(\sigma^2 + \sigma + 1) \simeq \mathbb{Z}_3[\zeta_3]$, a discrete valuation ring with uniformiser $\pi = 1 - \zeta_3$ and residue field $\mathbb{F}_3$. Being torsion-free of $\mathbb{Z}_3$ -rank 2, T is free of rank one over $\mathbb{Z}_3[\zeta_3]$, and $E[3] = T/3T \simeq \mathbb{Z}_3[\zeta_3]/(\pi^2)$ with σ acting as multiplication by $\zeta_3 = 1 - \pi$. Since $\pi \neq 0$ in $\mathbb{Z}_3[\zeta_3]/(\pi^2)$, this is a nontrivial unipotent automorphism of order 3; its fixed space is the line $\pi\mathbb{Z}_3[\zeta_3]/(\pi^2)$. The Serre conductor exponent at q is $\dim E[3] - \dim E[3]^{I_q} = 1$, there being no wild part. $\square$

Remark 2.3. For the other additive types the image of inertia mod 3 is not unipotent: it has order 2 or 4 (types I_0^* , III, III*), or it is minus a unipotent (types II, II*, defect 6) or a ramified quadratic character times a unipotent (type I_n^*). The last two are the twisted Steinberg shapes of Conjecture 1.4.

3. PROOF OF THEOREM 1.3

Let $f = f_E$, of level N with $q^2 \parallel N$, and $\bar{\rho} = \bar{\rho}_{E,3}$, which is irreducible by hypothesis and modular of weight 2 and level N . By Lemma 2.2 the Serre conductor $N(\bar{\rho})$ has exponent 1 at q and, as always, divides N elsewhere, so $N(\bar{\rho}) \mid N/q$. By the level-lowering theorem of Ribet, Carayol and Diamond [16, 4, 7], applicable at $\ell = 3$ under the hypothesis that $\bar{\rho}|_{\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))}$ is absolutely irreducible, there is a newform g of weight 2 and level $M \mid N(\bar{\rho})$, with coefficients in the ring of integers $\mathcal{O}$ of a number field, and a prime $\lambda \mid 3$ of $\mathcal{O}$ such that $a_p(g) \equiv a_p(f) \pmod{\lambda}$ for all primes $p \nmid 3N$. Since $M < N$, the forms $g(dz)$, $d \mid N/M$, are old at level N and lie in the orthogonal complement of f in $S_2(\Gamma_0(N))$. A congruence between the Hecke eigenvalues of f and of an eigenform in the orthogonal complement at almost all primes means that the Hecke algebra of level N has a maximal ideal of residue characteristic 3 containing the kernels of both eigencharacters, which is equivalent to 3 dividing the congruence number r_E of f [15], [1, §3]. Finally [1, Theorem 2.1] gives $\deg \varphi_E \mid r_E$ with $v_p(\deg \varphi_E) = v_p(r_E)$ for every prime p with $p^2 \nmid N$; as $9 \nmid N$, $v_3(\deg \varphi_E) = v_3(r_E) \geq 1$. $\square$

Remark 3.1. The hypothesis $9 \nmid N$ is used only in the last step, and it cannot simply be dropped from that step: Agashe, Ribet and Stein [1, Proposition 2.3] note that for the curve 99a1 the

prime 3 divides the congruence number but not the modular degree. The data nevertheless show the conclusion of the theorem for every curve with $9 \mid N$ and a prime of type IV or IV*; so do they at $q = 3$, where Lemma 2.2 does not apply because the ramification is wild. The proof gives one factor of 3 for each such prime but says nothing about additivity with the multiplicative contributions or with each other; that is the content of Conjecture 1.1 at $\ell = 3$.

4. THE COMPUTATIONS

4.1. Data and protocol. We used Cremona’s tables [6] in the form of the `ecdata` repository at commit `25cec5e`, which contain every elliptic curve over $\mathbb{Q}$ of conductor below 500000 with its Kodaira symbols, Tamagawa numbers, modular degree, isogeny class, torsion and, up to conductor 400000, the images of the mod ℓ Galois representations. Optimality is determined in the tables for every class of conductor at most 400000 and for 359,009 further classes up to 500000. The statements were found on the curves of conductor at most 300000, written down, and then tested once on the curves of conductor in $(300000, 400000]$, which had been held out, and afterwards on the extension to 500000. Nothing was adjusted after the hold-out test. The Kodaira types and discriminant exponents at the bad primes were recomputed with PARI/GP 2.15 [21] from the a -invariants, and fifteen modular degrees spanning the configurations were recomputed with `ellmoddegree`, all agreeing with the tables.

4.2. Results. Table 1 gives, for each range and each small prime ℓ , the number of optimal non-CM curves with $E[\ell]$ irreducible and a positive right-hand side in (1), the number of violations of Conjectures 1.1 and 1.4, and the number of Eisenstein curves with a positive bound together with the violations of Conjecture 1.5. There are none.

set	ℓ	curves	$E[\ell]$ irreducible, bound > 0	violations C1	violations C2	Eisenstein
working set, $N \leq 300000$	3	1,311,066	598,833	0	0	
	5	1,311,066	235,510	0		
	7	1,311,066	121,947	0		
	11	1,311,066	42,334	0		
	13	1,311,066	27,215	0		
hold-out, $300000 < N \leq 400000$	3	429,936	206,749	0	0	
	5	429,936	79,550	0		
	7	429,936	41,387	0		
	11	429,936	14,244	0		
	13	429,936	9,212	0		
extension, $400000 < N \leq 500000$	3	359,009	181,075	0	0	
	5	359,009	70,514	0		
	7	359,009	36,734	0		
	11	359,009	13,347	0		
	13	359,009	8,467	0		
out of table, $500000 < N \leq 4 \cdot 10^6$	3	154	110	0	0	

TABLE 1. Verification counts. “Bound > 0 ” means the right-hand side of (1) is positive.

The theorem range of Theorem 1.2 covers 281,739 of the 913,244 working-set pairs (E, ℓ) with $E[\ell]$ irreducible and a positive multiplicative bound; the remaining pairs have $\ell = 3$ (455,707), $\ell \mid N$ (175,798 at $\ell \geq 5$), or a contributing prime $q \equiv \pm 1 \pmod{\ell}$ (410,074), and show no violation either.

4.3. Isolated configurations. Table 2 isolates the additive types at $\ell = 3$: it lists, for curves with no multiplicative prime q with $3 \mid n_q$ and all additive primes of a single Kodaira type, the minimum of $v_3(\deg \varphi_E)$ by type and multiplicity. The hold-out reproduces every minimum.

type	one prime	two primes	three primes
IV	1 (25,414)	2 (969)	3 (11)
IV*	1 (30,168)	2 (2,268)	3 (45)
II	0 (34,611)	1 (2,073)	2 (24)
II*	0 (19,608)	1 (968)	2 (8)
III	0 (34,904)	0 (2,208)	0 (34)
III*	0 (25,574)	0 (1,307)	0 (29)
I ₀ *	0 (42,924)	0 (3,093)	0 (53)
I _n *	0 (133,322)	0 (28,125)	1 (1,483)

TABLE 2. Minimum of $v_3(\deg \varphi_E)$ (number of curves) over working-set curves whose additive primes are all of the given type, with no multiplicative 3-contribution and $E[3]$ irreducible.

The rows IV, IV* are Conjecture 1.1: each prime contributes exactly one at the minimum. The rows II, II* are Conjecture 1.4 with all weights equal to one. The row I_n* mixes primes with and without $3 \mid n$; separating them, two I_n* primes with $3 \mid n$ give 1 and three give 2. Across all configurations, the minimum of the excess over the bound of Conjecture 1.1 as a function of (number of II, II* primes, total I_n* weight) is exactly $\sum w_q - \max w_q$ in every cell with at least two contributing primes, and the “even subset” rule of Ribet–Takahashi type (which would predict 2 for two primes of weight 1) is violated 8,979 times.

4.4. Sharpness and the rest of the degree. The bounds are attained. Among working-set curves with $E[\ell]$ irreducible and a positive bound, $v_\ell(\deg \varphi_E)$ equals the bound of Conjectures 1.1 and 1.4 in 36.5% of cases at $\ell = 3$, 69.7% at $\ell = 5$ and 80.4% at $\ell = 7$, the proportion decreasing slowly with the conductor (from 75% to 35% at $\ell = 3$ between conductors 300 and 300000). The whole odd part of $\deg \varphi_E$ is another matter: for the 1,226,216 working-set optimal non-CM curves with no isogeny of odd degree, the odd part of $\deg \varphi_E$ equals $\prod \ell^{\text{bound}}$ in only 1.4% of cases. The quotient is most often 9, 3, 15, 45, 27 or 21; its smallest prime factor is 3 in two thirds of the cases and divides N in about half. So the local component groups account for a definite, sharp part of the modular degree and for little of the rest, which is governed by congruences of other origins and by the size of the Petersson norm.

4.5. Out of table. To test the statements beyond the tables, we generated curves with prescribed local types and conductor between 500000 and $4 \cdot 10^6$ (families with a type IV or IV* prime, with two or three type II primes, and random curves), kept those with trivial isogeny class (so that the curve is optimal and $E[\ell]$ is irreducible for every ℓ) and not CM, and computed their modular degrees with `ellmoddegree`. Table 3 summarises them; Conjectures 1.1 and 1.4 hold for all of them.

4.6. The Eisenstein case. Among the 75,430 working-set optimal non-CM curves with a rational 3-isogeny, 67,845 have a positive bound and 2,777 a positive deficit D ; $D = 2$ occurs 22 times. No curve without a rational 3-torsion point has a deficit, whatever its 3-isogeny structure. With a rational point of order 3 and mod 3 image of Borel type, $D \leq 1$; $D = 2$ occurs only when $E[3] \simeq \mathbb{Z}/3 \oplus \mu_3$ (the split Cartan image **3Cs.1.1** in the notation of [19], seventeen curves, all of rank 0 and torsion $\mathbb{Z}/3$) or when E has a point of order 9 (five curves). The bounds of Conjecture 1.5 hold with no exception on all three ranges, and the torsion bound is attained

family	curves	largest conductor	median seconds per degree
IV	63	3,971,200	0.9
IV*	45	3,994,600	4.9
IVmixed	45	3,977,584	1.2
random	1	1,151,077	77.3
all	154	3,994,600	1.5

TABLE 3. Curves outside the tables, with modular degrees computed independently.

with $D > 0$ in 2,773 cases. For $\ell = 5$ and 7 the deficits are at most 1 and occur only with a rational ℓ -torsion point (192 and 18 cases in the working set).

4.7. The twisted primes. [[TWIST]]

5. THE PRIME 2

At $\ell = 2$ the modular degree is the subject of Watkins’s conjecture $2^{\text{rank } E(\mathbb{Q})} \mid \deg \varphi_E$ [22] and of a substantial literature [8, 3, 23, 5, 9, 2], and the mod 2 representation has a rational 2-isogeny for a large fraction of curves. We record only the following. For working-set optimal curves without a rational 2-isogeny and not CM, the inequality $v_2(\deg \varphi_E) \geq \sum_{q \parallel N} v_2(n_q) + \#\{q : \text{type III, III}^*, \text{I}_0^* \text{ or } \text{I}_n^*\}$ holds with no exception on all three ranges (964,432, 325,510 and 313,741 curves), while counting I_0^* with weight $2 = v_2(\#\Phi_q^{\text{geom}})$ fails (123, 26 and 20 exceptions). The isolated-configuration minima suggest that types III, III* and I_n^* contribute 3 each and types II, IV and I_0^* contribute 1 each, in line with the theorem of Calegari and Emerton that a curve of odd modular degree has at most two odd bad primes and a rational 2-torsion point or prime conductor [3]. We have not attempted to reconcile the exponents with the results of [8, 5, 9, 2].

6. OPEN QUESTIONS

- (1) Prove Conjecture 1.1 at $\ell = 3$ for multiplicative primes, and remove the conditions $\ell \nmid N$ and $q \not\equiv \pm 1 \pmod{\ell}$ from Theorem 1.2. The data say the inequality survives both; whether the equality of Kim and Ota does is a separate question.
- (2) Prove the quantitative additive statement: a Kim–Ota type formula in which the prime q enters the level with exponent 2 and the local invariant is the unipotent ramification of Lemma 2.2.
- (3) Explain the “all but the largest” rule of Conjecture 1.4. The quadratic twist by the character ramified at two twisted primes turns both into untwisted types, which suggests comparing the congruence modules of f and of its twist, but the data exclude a congruence between f and its own twist (Section 4).
- (4) Identify the deficit of Conjecture 1.5 with a natural invariant of the Eisenstein ideal at ℓ , presumably through the cuspidal subgroup of $J_0(N)$ [12].

APPENDIX A. REPRODUCIBILITY

All scripts, the pinned data commit and the result tables are in the repository <https://github.com/dsr001001/mcptrial> (directory `ecmine`). The checker takes a table of optimal curves with their Kodaira symbols, Tamagawa numbers, modular degrees, isogeny degrees and torsion, and reports the violations of each statement; it runs on the full tables in minutes. A single curve can be checked in PARI/GP with

```
E = ellinit([0,0,0,-99,360]); L = ellglobalred(E)[5]; d = ellmoddegree(E);
b = sum(i=1,#L, my(k=L[i][2]); if(k>=5, valuation(k-4,3), if(k==4||k==-4, 1, 0)));
print([valuation(d,3), b])
```

where the codes $k = n + 4, 4, -4$ stand for I_n, IV, IV^* . The pipeline was rerun from a clean clone and reproduced every count.

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